

DISCRETE RANDOM VARIABLES

Chapter 2

WHAT IS A RANDOM VARIABLE?

- A random variable is one whose value cannot be predicted but is determined by the outcome of an experiment.

EXAMPLE

- two dice are rolled simultaneously a number of times. The sum of the numbers appearing uppermost is recorded. The possible outcomes we could expect are $\{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$.
- Since the possible outcomes may vary each time the dice are rolled, the sum of the numbers appearing uppermost is a random variable.

HOW DO WE EXPRESS A RANDOM VARIABLE?

- Random variables are expressed as capital letters, usually from the end of the alphabet (for example, X, Y, Z) and the value they can take on is represented by lowercase letters (for example, x, y, z respectively).

RANDOM VARIABLES

- The above situation illustrates an example of a discrete random variable since the possible outcomes were able to be counted. **Discrete random variables** generally deal with number or size.

- A random variable that can take on any value is defined as a **continuous random variable**. Continuous random variables generally deal with quantities that can be measured, such as mass, height or time.

EXAMPLE 1

Which of the following represent discrete random variables?

- a The number of goals scored at a football match
- b The height of students in a Maths Methods class
- c Shoe sizes
- d The number of girls in a five-child family
- e The time taken, in minutes, to run a distance of 10 kilometres

DISCRETE PROBABILITY DISTRIBUTIONS

- When dealing with random variables, the probabilities associated with them are often required.

EXAMPLE 2

- Let X represent the number of tails obtained in three tosses. Draw up a table that displays the values the discrete random variable can assume and the corresponding probabilities.

CONT.

- X denoted the random variable and x the value that the random variable could take. Thus the probability can be denoted by $p(x)$ or $\Pr(X = x)$. Hence the table in the example could be presented as shown below.

x	0	1	2	3
$\Pr(X = x)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

IMPORTANT CHARACTERISTICS THAT SATISFY ALL DISCRETE PROBABILITY DISTRIBUTIONS

- Each probability lies in a restricted interval $0 \leq \Pr(X = x) \leq 1$.
- The probabilities of a particular experiment sum to 1, that is,

$$\sum \Pr(x = x) = 1$$

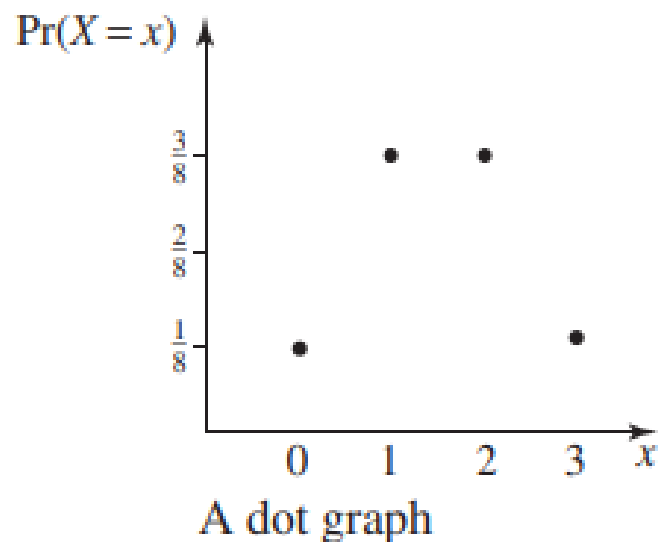
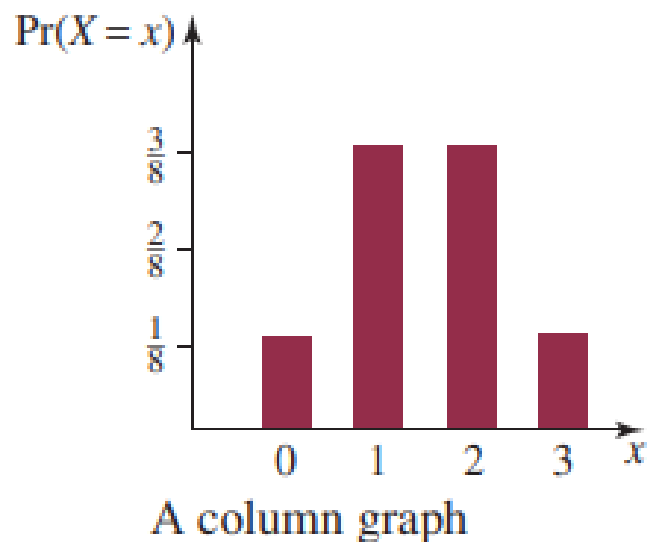
If these two characteristics are not satisfied, then there is no discrete probability distribution.

EXAMPLE 3

- Draw a probability distribution graph of the outcomes in worked example 12.

CONT.

Note: The probability distribution graph may also be drawn as follows.



EXAMPLE 4

Which of the following tables represent a discrete probability distribution?

a

x	0	1	2	3
$\Pr(X=x)$	0.2	0.5	0.2	0.1

c

x	-1	0	1	2
$\Pr(X=x)$	0.2	0.1	0.3	0.3

b

x	0	2	4	6
$\Pr(X=x)$	0.5	0.3	0.1	0.1

d

x	-2	0	5	7
$\Pr(X=x)$	-0.2	0.3	0.5	0.4

EXAMPLE 5

Find the value of k for each of the following discrete probability distributions.

a

x	1	3	5	7	9
$\Pr(X = x)$	0.2	k	0.2	0.3	0.1

b

x	0	1	2	3	4
$\Pr(X = x)$	$5k$	$6k$	$4k$	$3k$	$2k$

EXAMPLE 6

- a Show that the function $p(x) = \left(\frac{1}{42}\right)(5x + 3)$, where $x = 0, 1, 2, 3$ is a probability function.
- b Show that the function $p(x) = \left(\frac{1}{100}\right)(x^2)(6 - x)$, where $x = 2, 3, 4, 5$ is a probability function.

EXAMPLE 7

Three balls are selected from a box containing 6 blue balls and 4 yellow balls. If the ball chosen after each selection is replaced before the next selection, find:

a the probability distribution for the number of blue balls drawn:

i 0 blue balls

ii 1 blue ball

iii 2 blue balls

iv 3 blue balls

b the probability that 3 blue balls are chosen, given that at least one ball was blue.

MEASURES OF CENTRE OF DISCRETE RANDOM DISTRIBUTIONS

CONT.

- The expected value of a discrete random variable, X , is the average value of X . It is also referred to as the mean of X or the expectation.

EXPECTED VALUE OF A DISCRETE RANDOM VARIABLE, X

- The expected value of a discrete random variable, X , is denoted by $E(X)$ or the symbol μ (mu). It is defined as the sum of each value of X multiplied by its respective probability; that is,

$$\begin{aligned} E(X) &= x_1 \Pr(X = x_1) + x_2 \Pr(X = x_2) + x_3 \Pr(X = x_3) + \dots + x_n \Pr(X = x_n) \\ &= \sum_{\text{all } x} x \Pr(X = x) \end{aligned}$$

- The expected value will not always assume a discrete value.

EXAMPLE 8

- Find the expected value of a random variable that has the following probability distribution.

x	1	2	3	4	5
$\Pr(X = x)$	$\frac{2}{5}$	$\frac{1}{10}$	$\frac{3}{10}$	$\frac{1}{10}$	$\frac{1}{10}$

EXAMPLE 9

- Find the unknown probability, a , and hence determine the expected value of a random variable that has the following probability distribution.

x	2	4	6	8	10
$\Pr(X = x)$	0.2	0.4	a	0.1	0.1

EXAMPLE 10

- Find the values of a and b of the following probability distribution if $E(X) = 4.29$.

x	1	2	3	4	5	6	7
$\Pr(X=x)$	0.1	0.1	a	0.3	0.2	b	0.2

EXAMPLE 11

Niki and Melanie devise a gambling game based on tossing three coins simultaneously. If three heads or three tails are obtained, the player wins \$20. Otherwise the player loses \$5. In order to make a profit they charge each person \$2 to play.

- a What is the expected gain to the player?
- b Do Niki and Melanie make a profit?
- c Is this a fair game?

EXPECTATION THEOREMS

EXAMPLE 12

- A random variable has the following probability distribution.

x	1	2	3	4
$\Pr(X = x)$	0.25	0.26	0.14	0.35

- Find: a $E(X)$ b $E(3X)$ c $E(2X - 4)$ d $E(X^2)$.

SUMMARY

- $E(aX) = aE(X)$ where X is a random variable and a is a constant.
- $E(aX + b) = aE(X) + b$ where X is a random variable and a and b are constants.
- $E(b) = b$ where b is a constant.
- $E(X + Y) = E(X) + E(Y)$ where X and Y are both random variables
- $E(X^2) \neq [E(X)]^2$

EXAMPLE 13

- Casey decides to apply for a job selling mobile phones. She receives a base salary of \$200 per month and \$15 for every mobile phone sold. The following table shows the probability of a particular number of mobile phones, x , being sold per month. What would be the expected salary Casey would receive each month?

MEDIAN AND MODE

- The median is the middle value of the distribution. It is the value such that 50% of the distribution lies to the left of this value, and 50% of the distribution lies to the right.
- For a random variable, X , the mode is the most commonly occurring value, that is, it is the variable with the highest probability.

EXAMPLE 14

- For the following probability distributions, calculate:
i the median ii the mode.

a

x	3	5	6	8
$\Pr(X = x)$	0.25	0.4	0.15	0.2

b

x	5	8	11	14
$\Pr(X = x)$	0.18	0.32	0.2	0.3